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MODULE 1

ELECTROCHEMISTRY AND CORROSION

Electrochemistry

Electro Chemistry deals with the chemical reactions produced by passing electric current through an electrolyte or the production of electric current through chemical reactions.

Differences between Electrochemical cells & Electrolytic cells

Electrochemical cells	**Electrolytic cells**
A device which converts chemical energy to electrical energy	**A device which converts electrical energy to chemical energy.**
Consists of two electrodes immersed in one or more suitable electrolyte and the electrodes are connected externally	Both the electrodes are inserted in the same electrolyte solution.
Anode is -ve **(LOAN)**	Anode is +ve
Free energy change (ΔG) is-ve (Spontaneous)	Free energy change (ΔG) is +ve
Salt bridge is used	Salt bridge is not used.
Eg : Daniel Cell	Eg : Electrolysis

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Daniel Cell

A Daniel cell is the best example of a galvanic cell which converts chemical energy into electrical energy. It consists of two electrodes of dissimilar metals, Zn and Cu; each electrode is in contact with a solution of its own ion; Zinc sulphate and copper sulphate respectively.

The Daniel cell can be conventionally represented as $Zn/Zn^{2+}|| Cu^{2+}/Cu$

- In a Daniel cell electrons flow from zinc electrode to copper electrode through an external circuit, while metal ions form one half cell to the other through the salt bridge.
- Here current flows from copper electrode to zinc electrode that is cathode to anode via an external circuit.
- Daniel cell is a reversible cell

Cell Reactions

Ion $Zn/ZnSO_4$ half-cell, oxidation reaction occurs. (anode)

$Zn \rightarrow Zn^{2+} + 2e^-$

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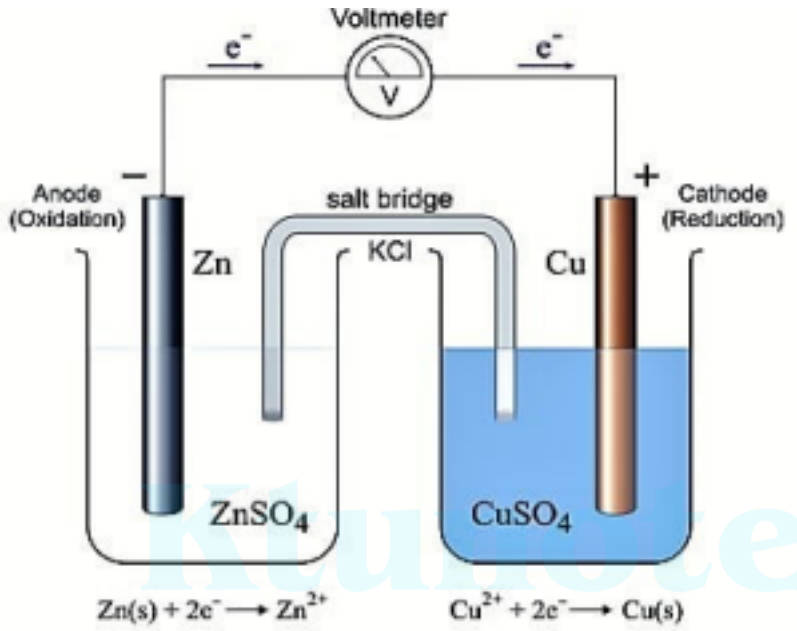
Ion Cu/CuSO4 half-cell, reduction reaction occurs. (cathode)

$Cu^{2+} + 2e^- \rightarrow Cu$

The net cell reaction is

$Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$

Cell is represented as $Zn/Zn^{2+}|| Cu^{2+}/Cu$



$Zn(s) \mid ZnSO_4(aq) || CuSO_4(aq) \mid Cu(s)$

SALT BRIDGE: Salt bridge is an inverted U tube. Containing inert electrolytes salts like KCl or NH4NO3 in agar-agar or gelatine. The ions of K+ and Cl- have equal mobility.

Functions of Salt bridge

- A salt bridge maintains the electrical neutrality of each half cells.
- It prevents the accumulation of charges.
- Salt bridge also prevents the diffusion of interfering ions from one compartment to another.
- A salt bridge minimizes or eliminates the liquid junction potential.
- Salts like KCl, NH4NO3 are used for making salt bridge. Ions like K + and Cl - has equal mobility and get drifted to cathode and anode compartments respectively under the potential difference.

Single Electrode Potential

It is measure of tendency of an electrode to lose or gain electron. Tendency of electrode for

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oxidation (loose electrons) is oxidation potential and tendency for reduction (gain electrons) is reduction potential Ered. Ered = - Eoxd

Factors affecting electrode potential

1.Concentration of metal irons in the solution

2.Temperature of the solution

3.Nature of the metal

REVERSIBLE AND IRREVERSIBLE CELLS

A cell is said to be reversible if it satisfies the following conditions ,

1) If an emf exactly equal to that of a cell is applied the chemical reaction taking place in the cell will stops

2)If the external emf is slightly greater than the actual emf of the cell applied, the cell reaction gets reversed.

Eg : Daniel cell

If a cell does not obey the following conditions it is said to be irreversible. Eg : Dry cell

NERNST EQUATION

The relation between the electrode potential and the concentration of the electrolyte solution is given by Nernst equation.

Walter Nernst derived a mathematical relationship which enables us to calculate **the effect of electrolyte concentration on electrode potential.**

For a half cell,

$$E=E^0 - 2.303RT / nF \log 1/ [M^{n+}]$$

$$E=E^0 - 0.0591/ n \log 1/[M^{n+}] \text{ at } 25^0C$$

where, E^0=Standard electrode potential R=gas constant; T= temperature in K n=number of electrons transferred in the half cell reaction; F=faraday of electricity; $[M^{n+}]$= concentration of electrolyte.

DERIVATION

Nernst equation gives us the effect of electrolyte concentration on electrode potential.

Consider a general reduction reaction, taking place at the electrode $M^{n+} + ne \rightarrow M$

The decrease in free energy, ΔG accompanying this process is given by the well-known thermodynamic equation (Vant Hoff's equation)

$$\Delta G=\Delta G^0 + RT \ln Q \quad .... (1)$$

R = gas constant , Q is the reaction quotient = Conc of Products/ conc of reactant [P]/[R]

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In a reversible cell, electrical energy is produced at the expense of decrease in free energy, and free energy is related with EMF of the cell as follows

$$\Delta G=-nFE, \quad \Delta G^0=-nFE^0$$

Substitute in (1) $-nFE=-nFE^0+ RT \ln Q$

Dividing by –nF

$$E= E^0 - RT / nF \ \ln Q \ldots (2)$$

Substituting the values of constants at the temperature T = 298 K, R= 8.314JK-1 mol^{-1}, F=96500C, Q=[M] / [M^{n+}] **and converting in to natural log**

$$E= E^0 - 2.303RT/nF \log [M] / [M^+]$$

Pure solid [M] is taken as one $E = E^0 - 2.303RT/ nF \log 1 / [[M^{n+}]$

This equation is called Nernst equation. It's clear that electrode potential is directly proportional to molar concentration of ions in the same solution and temperature.

Nernst Equation for a complete cell

For a cell reaction a A + b B $\longrightarrow$ c C+ d D The reaction quotient Q can be written as **Q**

$$= [C]^c[D]^d / [A]^a[B]^b$$

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Thus the Nernst equation for a complete cell is given by

$$E_{cell} = E^0_{cell} - 0.0591/n \log Q$$

$$E_{cell} = E^0_{cell} - 0.0591/n \log [C]^c[D]^d / [A]^a[B]^b$$

For a Daniel cell , $Cu^{2+} + Zn \longrightarrow Zn^{2+} + Cu$ $Q = [Zn^{2+}] [Cu] / [Cu^{2+}][Zn]$ $E_{cell} = E^0_{cell} - 0.0591/n \log Q$

Here concentration of pure solids are taken as one then,

$Q = [Zn^{2+}]/ Cu^{2+}]$ $\quad E_{cell} = E^0_{cell} - 2.303RT/nF \log [Zn^{2+}]/ Cu^{2+}]$

$$E_{cell} = E^0_{cell} - 0.0591/n \log [Zn^{2+}]/ Cu^{2+}]$$

free energy and EMF: The cell in a reversible cell, electrical energy is produced at the expense of decrease in free energyΔG,

Electrical energy produced per mol= nFE

$\Delta G=-nFE$ where F is called Faraday which is the charge of one mole of electrons. (F=6.023x1023x1.6x10-19=96500 C); n is number of electrons; E is EMF

Variation of EMF of Hydrogen Electrode with pH of a solution

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For a Hydrogen Electrode at 25^0C $H^+ + 1e \longrightarrow ½ H_2$

Nernst equation can be written as,

$$E = E^0 + 0.0591/n \log [H^+] \quad pH = -\log[H^+]$$

$$E = -0.0591pH$$

Case 1: When pH = 0 i.e. E= -0.0591*0= 0 V (acidic condition)

Case 2: When pH = 7 i.e. E= - 0.0591*7 = - 0.414V(Neutral)

Case 3: When pH =14 i.e. E= -0.0591*14=-0.825 V (Alkaline)

Variation of EMF of Oxygen Electrode with pH of a solution

For oxygen electrode at 25^0C $O_2 + 4H^+ + 4e \longrightarrow 2 H_2O$ Nernst equation can be written as,

$$E = E^0 + 0.0591/n \log [H^+]^4 \quad n=4 \quad E^0 = 1.23V$$

$$E = E^0 + 0.0591/4 \log [H^+] *4$$

$$E = 1.23 - 0.0591 \log pH$$

Case 1: When pH = 0 i.e. E=1.23-0.0591*0= 1.23V

Case 2: When pH = 7 i.e. E=1,23-0.0591*7= 0.81V

Case 3: When pH =14 i.e. E=1.23-0.0591*14=0.40V

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Applications of Nernst equation

1. To study the effect of electrolyte concentration on electrode potential.

2. Calculation of equilibrium constant of a reaction.

At equilibrium, anodic and cathodic reactions become equal. So E_{cell} become 0.

So the Nernst equation is, $E_{cell} = E^0_{cell} - 0.0591/n \log Q$ (at equilibrium Q= K the equilibrium constant)

So at equilibrium , $0= E^0_{cell} - 0.0591/n \log K$, $E^0_{cell} = (0.0591/n) \log K$ Log K = $n E^0cell /0.0591$

3. Determination of EMF of a Cell

4. Determination of unknown concentration.

5. Determination of a p H of a solution.

6. Used for finding out valency of an ion

NUMERICALS

1. The standard electrode potential of zinc ions is 0.76V. What will be the potential of a 2M solution at 300K?

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The Nernst equation for the given conditions can be written as follows;

$EM^{n+}/M = E^o + [(2.303RT)/nF] \times \log [Mn^+]$

Here, E° = 0.76V n = 2 F = 96500 C/mole $[Mn^+]$ = 2 M R =8.314 J/K mole T =300 K

Substituting the given values in Nernst equation we get,

$EZn^{2+}/Zn = 0.76 + [(2.303\times8.314\times300)/(2\times96500)] \times \log 2 = 0.76 + [0.0298 \times (0.301)] =$

$0.76 + 0.009 =$ **0.769V**

Therefore, the potential of a 2M solution at 300K is 0.769V.

2.What is the Cell Potential of the electrochemical cell in which the cell reaction is: $Pb^{2+} + Cd \rightarrow Pb + Cd^{2+}$; Given that E^o_{cell} = 0.277 volts, temperature = 25°C, $[Cd^{2+}]$ = 0.02M, and $[Pb^{2+}]$ = 0.2M.

Since the temperature is equal to 25°C, the Nernst equation can be written as follows;

$E_{cell} = E^0_{cell} - (0.0592/n) \log_{10}Q$

Here, two moles of electrons are transferred in the reaction. Therefore, n = 2. The reaction quotient (Q) is given by $[Cd^{2+}]/[Pb^{2+}]$ = (0.02M)/(0.2M) = 0.1.

The equation can now be rewritten as:

$E_{cell} = 0.277 - (0.0592/2) \times \log_{10}(0.1) = 0.277 - (0.0296)(-1) =$ **0.3066 Volts** Thus, the cell potential of this electrochemical cell at a temperature of 25°C is 0.3066 volts.

3.The Cu^{2+}ion concentration in a copper-silver electrochemical cell is 0.1M. If $E^o(Ag^+/Ag)$ = 0.8V, $E^o(Cu^{2+}/Cu)$ = 0.34V, and Cell potential (at 25°C) = 0.422V, find the silver ion concentration.

Here, the silver electrode acts as a cathode whereas the copper electrode serves as the anode. This is because the standard electrode potential of the silver electrode is greater than that of the copper electrode. The standard electrode potential of the cell can now be calculated, as shown below.

$E^o_{cell} = E^o_{cathode} - E^o_{anode} = 0.8V - 0.34V = 0.46V$

Since the charge on the copper ion is +2 and the charge on the silver ion is +1, the balanced cell reaction is:

$Cu \longrightarrow Cu^{2+} + 2e$

$2Ag^+ + 2e \longrightarrow 2Ag$

$2Ag^+ + Cu \rightarrow 2Ag + Cu^{2+}$

Since two electrons are transferred in the cell reaction, n = 2. Now, the Nernst equation for this electrochemical cell can be written as follows.

$E_{cell} = E^0_{cell} - (0.0592/2) \times \log (0.1/[Ag^+]^2)$

$0.422V = 0.46 - 0.0296 \times (-1 - 2\log [Ag^+])$

Therefore, $-2\log [Ag^+] = 1.283 + 1 = 2.283$

Or, $\log [Ag^+] = -1.141$

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$[Ag^+]$ = antilog (-1.141) = **0.0722 M**

4. Find the cell potential of a galvanic cell based on the following reduction half-reactions at 25 °C

$Cd^{2+} + 2e^- \rightarrow Cd$ $E^0 = -0.403$ V

$Pb^{2+} + 2e^- \rightarrow Pb$ $E^0 = -0.126$ V

where $[Cd^{2+}] = 0.020$ M and $[Pb^{2+}] = 0.200$ M.

5. **Calculate the emf of the cell. Zn|Zn2+(0.001M)||Cu2+(0.1M)|Cu The standard potential of Cu/Cu2+ half-cell is +0.34 and Zn/Zn2+ is -0.76 V.**

Given Zn|Zn2+(0.001M)||Cu2+(0.1M)|Cu

Overall cell reaction:

Zn⟶Zn2++2e− Cu^{2+}+2e−⟶Cu

Zn+Cu2+⟶Zn2++Cu

E^0_{cell}=standard reduction potential of cathode - standard oxidation potential of anode

E^0cell=0.34 +0.76 V

E^0cell=1.1 V

K_C=[Cu2+]/[Zn2+]=10−1/10−3=10^2

EMF of the cell at any electrode concentration is:

E=Eo−0.059/n log (K_c)=1.1− 0.0295log(10^2)=1.1-0.02951×(2)=1.1−0.059=1.041V

6. What will be the standard electrode potential of Ni2+ / Ni electrode if the cell potential of the cell Ni / Ni 2+ (0.1M) // Cu2+(0.01M) / Cu is 0.59 V at 25oC? E0 Cu 2+ /Cu = 0.34 V. E cell = E0cell – (0.0591/n) log [Ni 2+] / [Cu2+]

0.59 = E0cell - (0.0591/2) log 0.1 / 0.01, E0cell = 0.59 + 0.02955 = 0.61955 V

E0cell = E0cathode - E0anode E0cell = E0Cu2+ /Cu – E0 Ni2+ /Ni

0.61955 = +0.34 – E0 Ni2+ /Ni

E 0 Ni2+ /Ni = - 0.27955 V

At anode Ni (s) → Ni2+ + 2e

At cathode Cu2+ + 2e → Cu(s)

Electrochemical Series

A series which represents an arrangement of various metal electrodes in the increasing order of their standard reduction potentials or decreasing order of oxidation potential is called electrochemical series.

Li, K, Na, Mg, Al, Zn, Fe, Ni, H_2, Ag, Hg, Cu, Au, F (Increasing order of reduction potential)

Applications of Electrochemical Series

1. **Reactivity of Metals**:

Electrodes having negative values of standard electrode potential with respect to hydrogen

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have tendency to undergo oxidation and act as anode. whereas electrodes having positive values have tendency to undergo reduction and act as cathode in aqueous solution.

2. Displacement reaction:

a) To predict whether a given metal will displace another from its salt solution.

Metals having lower reduction potential will always displace metals having higher reduction potential.

E.g.: Zn metal can displace Cu from CuSO4

b) Metals having negative electrode potentials can displace hydrogen gas from acids.

e.g.: Metals like Fe, Zn, Al etc. Can liberate H2 gas when react with acids.

3.Spontaneity of a redox reaction:

The emf of the cell is positive, the cell reaction is spontaneous. If emf of a cell is negative, the reaction will not occur at all.

4. Calculation of EMF of a cell:

EMF of the cell can be readily calculated from the relation

$E_{cell} = E^0_R - E^0_L$

5.Predicting the anode and cathode:

The electrode with higher reduction potential will act as cathode and the electrode having lower electrode potential will act as anode

6. Prediction of relative oxidising and reducing tendency.

7.Corrosion:

The metal higher in the series is anodic or more active and they are more prone to corrosion. The metals lower in the series are noble metals and they are less prone to corrosion.

8. Predicting the Product of Electrolysis.

Problems

1. Write the product of electrolysis when we electrolyse the acidified mixture of NaCl and KI in aqueous solution? Given that E^0Cl_2/Cl^- = 1.36 V E^0I2/I^-=0.54V E^0K^+/K^- =-2.93V E^0Na+/Na= -2.71

During electrolysis the cell behaves like an electrolytic cell. So anode is positive

The ions going to anode are Cl^-, I^- From electrochemical series the electrode potential of $I^-$is less than the electrode potential of Cl so I^- goes to anode and the reaction is

At anode (Oxidation) $2I \longrightarrow I_2 + 2e$

Cathode is positive and the ions goes to cathode are H^+,Na^+ & K^+ The electrode potential of $H^+$is higher than Na^+ & $K^+$the H+ goes in to cathode.

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At Cathode (Reduction) $2H^{+}+2e$ ═══ H_2

REFERENCE ELECTRODES

A reference electrode is an electrode which has a stable electrode potential with which we can compare the electrode potential of other electrodes.

Reference electrodes are of two types. Primary reference electrode and secondary reference electrode.

Primary reference electrode: Electrodes whose electrode potential is taken as zero at all temperatures**. Eg: Standard Hydrogen Electrode (SHE**)

Standard Hydrogen Electrode (SHE)

The standard hydrogen electrode is a gas electrode, used as primary reference electrode..Its electrode potential is taken as 0.

Construction:

- It is a primary reference electrode.
- A small Pt plate coated with Pt black is connected to a Pt wire which is sealed in a glass jacket.
- Glass jacket having an inlet for Hydrogen gas (at 1atm pressure and 298K) and a number of small holes at the bottom for the escape of Hydrogen gas
- It is dipped in 1M HCl solution at 298K.

Working

• SHE is a reversible electrode which can act as cathode or anode, depending on the potential of electrode to which it is coupled.
If it act as anode the electrode can be represented as, The electrode can be represented as Pt,H2 (g)1atm/, H^{+}(1M)
If it act as cathode the electrode can be represented as, H^{+}(1M)/ H2 (g)1atm,Pt

• At the cathode,
2H+ + 2e −→H2(g) (reduction), If the potential of coupled electrode is less than zero (Eg: Zn, Mg)

• At the anode,
H2(g)→2H + + 2e − (oxidation) If the potential of coupled electrode is greater than zero (Eg:CU, Ag)

Limitations

• It is readily affected by compounds of Hg, As and S and it cannot be used in

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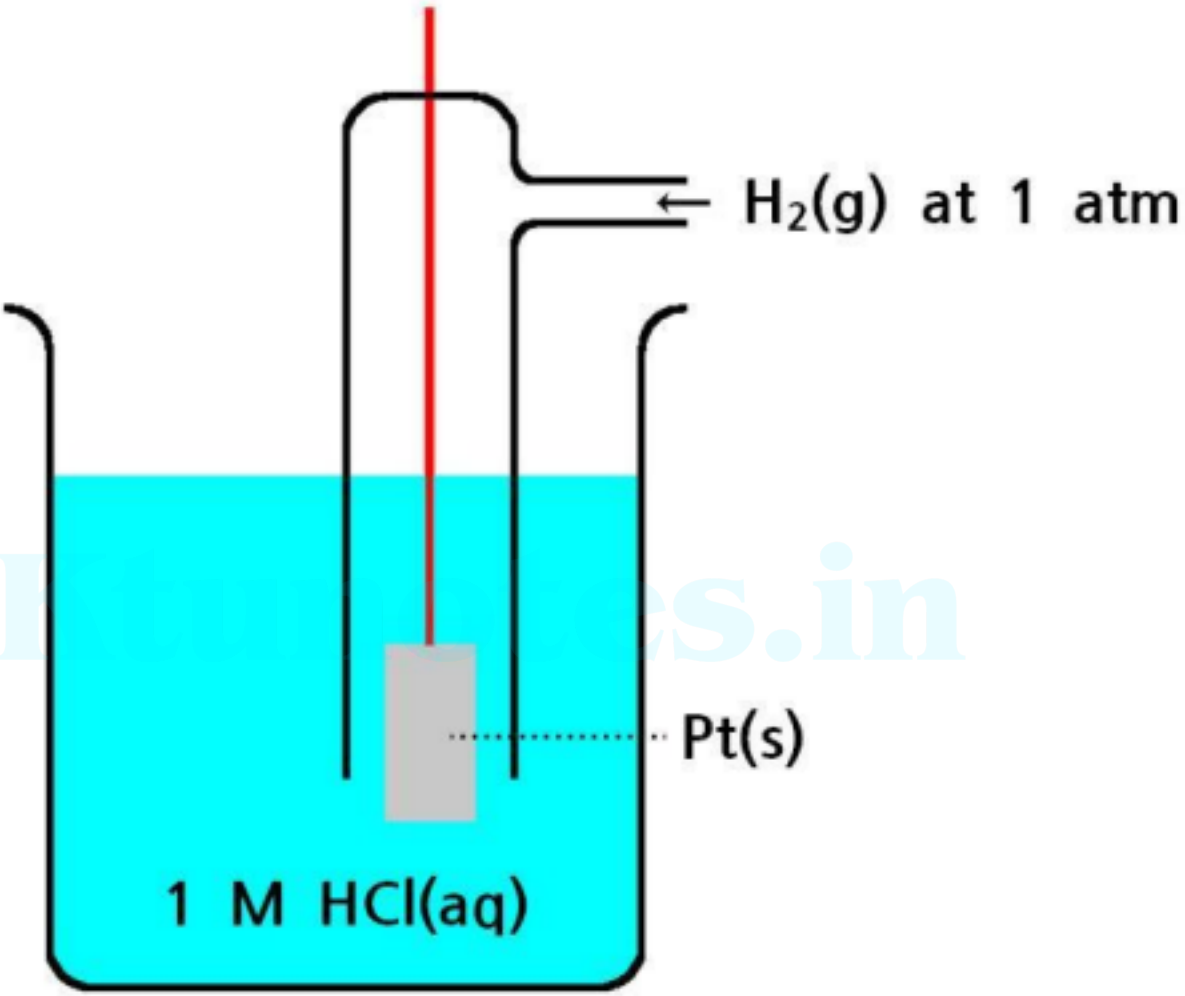
solution Containing these ions.

• It cannot be used in presence of strong oxidizing and reducing agents.

• It is expensive due to Pt.

• The Pt surface must be specially prepared and maintained or it is difficult to maintain Pt surface.



Applications

• Used as a reference electrode for increasing the electrode potential of other electrodes. • Used in the determination of pH

• it can be used over the entire pH range

Secondary reference electrode: Electrodes with known or constant electrode potential

E.g.: Calomel electrode, glass electrode, quinhydrone electrode

CALOMEL ELECTRODE

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Saturated calomel electrode is a secondary reference electrode. This is an example of metal -metal insoluble metal salt electrode and solution of its common ion.

Construction

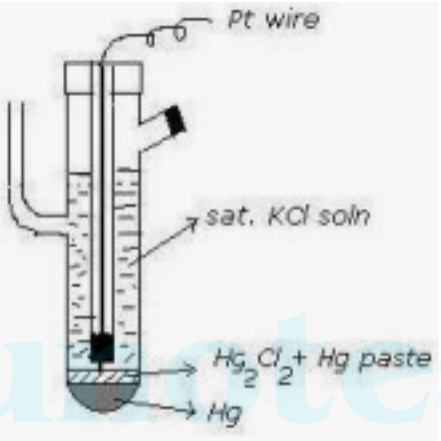
• It consists of a glass tube in the bottom of which is a layer of Hg.

• Over the Hg layer, there is a layer of solid mercurous chloride. (Hg2Cl2) •

The remaining portion is filled with normal, decinormal or saturated KCl • A

Pt wire is dipped in to the Hg- layer is used for making electrical contact. • The

side tube is used for making contact with a salt bridge.



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Working

It can act as an anode or cathode depending on the electrode potential of the coupled electrode.

• If it can act as anode,

$2Hg + 2Cl^- \rightarrow Hg_2Cl_2 + 2e$ Here the electrode potential of the coupled electrode is greater than calomel electrode. Can be represented as, Hg / Hg_2Cl_2, KCl •

If it can act as cathode,

$Hg_2Cl_2 + 2e^- \rightarrow 2Hg + 2Cl^-$ Here the electrode potential of the coupled electrode is less than that of calomel electrode. Can be represented as KCl, Hg_2Cl_2/ Hg

Applying Nernst Equation to reduction reaction,

$$E_{calomel} = E^0_{calomel} - 2.303RT / nF \log [Hg]^2[Cl^-]^2 / [Hg_2Cl_2]$$

For solids concentration is taken as 1

$$E_{calomel} = E^0_{calomel} - 2.303RT / 2F \log [Cl^-]^2$$

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$$E_{calomel} = E^0_{calomel} - 2.303RT / F \log[Cl^-]$$

From the above equation it is seen that the potential of the calomel electrode varies with the concentration of the KCl solution used,

Electrode potential of Calomel electrode at different concentrations are

KCl concentration	0.1N (DNCE)	1N (NCE)	Saturated (SCE)
Electrode potential in Volt	0.335V	0.2810V	0.2422V

Applications

1. Used for determination of electrode potential.
2. Used to find out pH of an unknown solution.

Limitations

1. Toxic nature of mercury
2. cannot be used above 500 c because the mercurous chloride breakdowns yielding an unstable reading.

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GLASS ELECTRODE

Glass electrode is a secondary reference electrode which produces a constant and reproducible electrode potential. It is ion selective electrodes and used for the pH measurements.

Construction and Working •

It is made up of a special type of glass relatively low melting point and high electrical conductivity

Long glass tube with a thin glass bulb filled with 0.1M HCl.

A Pt wire or silver wire coated with a layer of AgCl is inserted into it to make electrical contact.

Composition: SiO_2 = 72%, CaO= 6%, Na_2O= 22%

Electrode representation is, **Pt/0.1MHCl, glass/Test H+ solution**

The electrode potential of glass electrode is

$$E_G = E_G^0 + 0.0591 \log [H+]$$

E_G= **EG ^{0}- 0.0591[pH]** E_G^0 is a constant depending upon nature of the glass and pH of the solution taken inside the glass tube .E_G^0 **is 0.6V or 0.5V depending om the variety of the**

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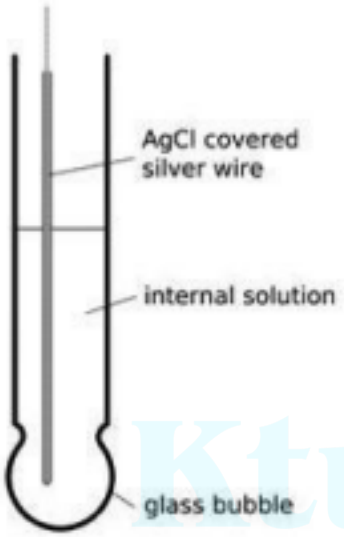
class

Advantages

1. It is simple and easily be used.
2. Used in pH range 0 -12.
3. The results are accurate.
4. Can be used in presence of strong oxidising and reducing agents.
5. It can be used in coloured, turbid and colloidal solutions



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Determination of pH by using glass electrode

Principle:

The H+ ion concentration inside the electrode is constant. When this electrode is dipped in to solution of unknown H+, it becomes sensitive to the outside concentration in the solution. The potential difference is **developed** across the membrane. By measuring that potential difference, we can find out pH.

Procedure

In order to measure the potential of glass electrode the unknown solution is taken in beaker, Glass electrode is dipped in it. It is connected to a saturated calomel electrode. emf of the cell is measured. The cell formed is represented as ,

$$\mathbf{Hg / Hg_2Cl_2\ KCl//\ [H^+]_{(c=?)} /Glass/\ H^+_{(0.1M)}, Pt,}$$

$$\mathbf{E_{cell}=E_C-E_A}$$

$$E_{cell}= E_{glass} - E_{calomel}$$

$$E_{cell} = [E^0_G - 0.0591pH] - 0.2422$$

$$E_{cell} =[E^0_G - 0.0591pH]- 0.2422$$

$$E_{cell} =(E^0_G - 0.2422)- 0.0591pH$$

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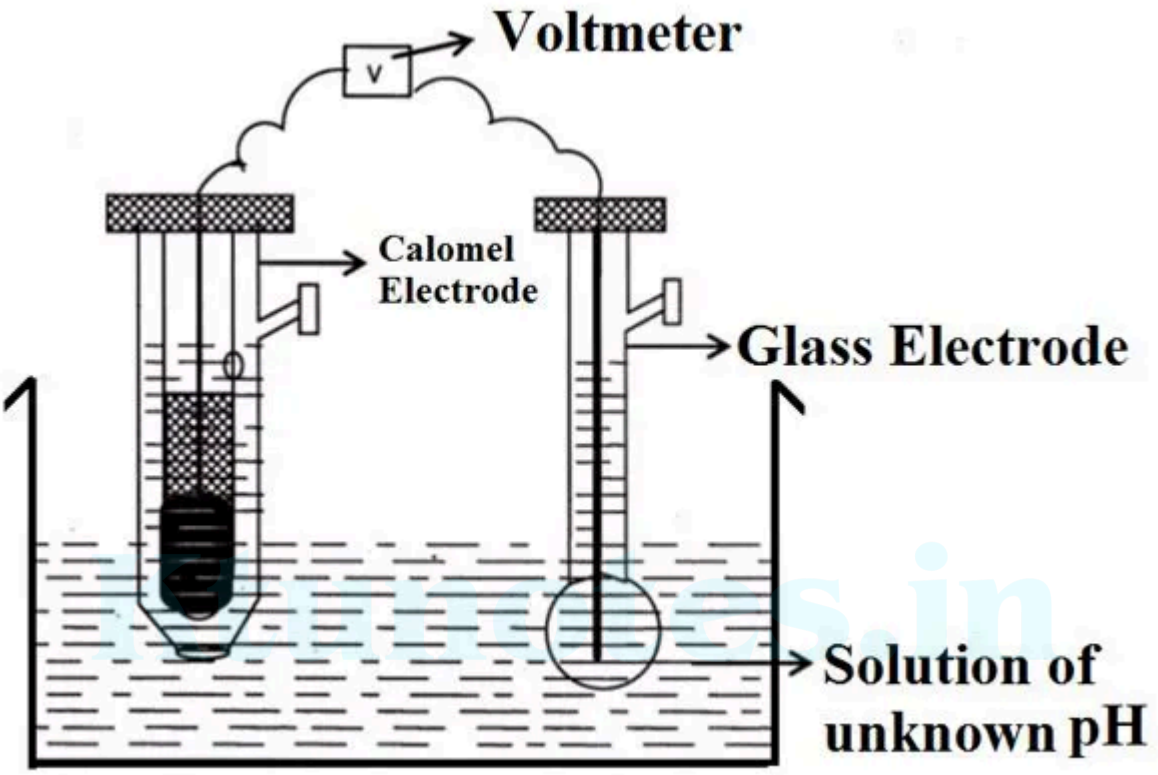
This equation has the form Y= C+mX the equation of a straight line having Y intercept C and slope m,when we plot a graph between emf on the Y-axis and pH on the X-axis.

The constants C and m can be determined by calibrating the pHmeter with two buffer solutions of known pH.The intercept C= E^0_G - 0.2422 and the slope is -0.0591 at 25^0C. The pH of the test solution can be calculated by noting the Ecell with test solution and knowing the values of C and m.



Determination of pH using glass electrode

Limitations

1. As the glass electrode has very high electrical resistance, electronic potentiometer must be used to find EMF.

2. Due to highly absorptive nature a lot of working is required

1)A pH buffer solution shows an emf of 208 mV with pH 4 buffer solution and -32 mV with pH 9.2 buffer solution. Find the pH of test solution if it shows an emf of 78 mV.

Ans: The equation has the form y=c+mx. The equation for a straight line having intercept c and slope m when we plot a graph between emf (y-axis) and pH (x-axis).

208 = c + 4m (1)

-32 = c + 9.2m (2)

equation (1)-(2) gives

240 = -5.2m m = -46.15

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substituting in (1)

208 = c + 4 x -46.15 c = 392.6 Emf of the test solution is

78 = c*m*pH Knowing c and m unknown pH can be determined by substituting .

78= 392.6+-46.15 *pH pH = 6.81

TYPES OF CELLS

Commercial cells are used as a source of electrical energy.

There are of three types.

1. **Primary cells**- In which cell reactions cannot be reversed. So, it cannot be recharged.

Eg: Daniel cell, Weston cadmium cell

2.**Secondary cells** (storage or accumulation cell)- In which cell reaction can be reversed by applying an external source of current.

Eg: Lithium-ion cell, Lead storage battery, Ni-Cd battery.

LITHIUM-ION CELL

• It is a rechargeable battery in which Li ion moves between anode and cathode during charging and discharging.

It is more popular than other batteries because, Li has higher negative value for reduction potential and a lower atomic mass than any other metal.

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CONSTRUCTION

In a typical Li ion cell , The positive electrode is made up of crystalline Li metal oxide (LiMO2) Eg : $LiCoO_2$ or $LiMnO_2$..

The negative electrode is made up of lithiated graphite or lithiated carbon (Li_XC_G) Lithium ions are inserted in between the layer of carbon atoms.

Electrolyte is the solution of lithium salt in an organic solvent.($LiBF_4$)

A solid electrolyte separator (SEI)separates the anode and cathode.

WORKING

The lithium ion moves from the anode to cathode during discharging and from cathode to anode while charging.

Charging

During Charging the cell act as an electrolytic cell the electrical energy is converted in to chemical energy.

Anode (+)

$$LiMO_2 \rightarrow Li(1-x)\ MO_2 + xLi^+ + xe$$

Cathode (-)

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$$C_G + xLi^+ + xe^- \rightarrow Li_XC_G$$

The net cell Reactions is,

$$LiMO_2 + C_G \rightarrow Li_XC_G + Li(1-X)MO_2$$

Discharging:

During discharging the cell act as an electrochemical cell where the chemical energy is converted to electrical energy. When the cell acts as an electrochemical cell, LiMO2 is the cathode and graphite is the anode.

Anode (-)

$Li_XC_G \rightarrow xLi^+ + xe^- + C_G$ (Oxidation) These Li + ions produced during oxidation diffuses through the electrolyte and reaches at the cathode for reduction.

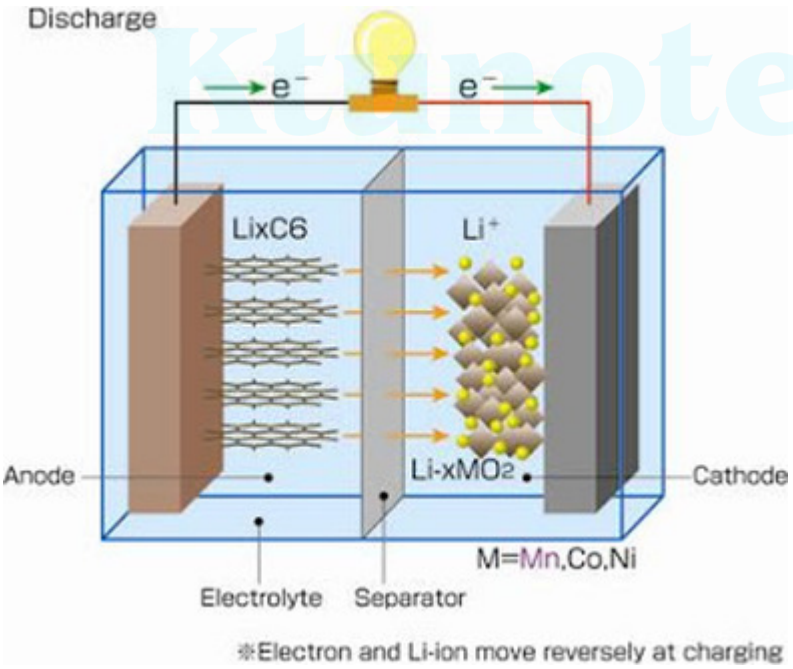
Cathode (+)

$$Li(1-x)MO_2 + xLi^+ + xe^- \rightarrow LiMO_2$$

The net cell Reactions is,

$$LixC_G + Li(1-x)MO_2 \rightarrow LiMO_2 + C_G$$

During discharging the Li cell is represented as,



Advantages

- Li-ion batteries are lighter than other equivalent batteries.
- Can handle hundreds of charge discharge cycles
- They do not use poisonous materials like Pb, Hg,Cd etc

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• Available in different size and shape.

• They have no memory effect-which means that if do not have completely discharge them before recharging.

Disadvantages

• They will be destroyed 2 or 3 years from the date of manufacture whether you used them or not.

• They are extremely sensitive to high temperature.

• Expensive to manufacture.

• If they get completely discharged, then destroyed

Applications

• Used in notebook computers • Mobile phone and other electronic devices. • Used in small digital music devices, MP3 player. • Digital camera. • Laptops. • Military applications.

Li ion cell cannot be charged 100% because when battery is charged to 100 the reaction becomes, $LiMO_2 \rightarrow Li(1-1)\ MO_2 + xLi^+ + xe-$ i.e. $LiMO_2 \rightarrow LiMO_2 + xLi^+ + xe$ metal oxides are highly explosive. Hence automatic circuit breakers in Li-ion battery.

Fuel Cells:

A fuel cell is a galvanic cell that converts chemical energy of a fuel and an oxidant directly into electrical energy.A fuel cell can be generally represented as Fuel/-ve electrode/electrolyte/+ve electrode/oxidant.
It is an electrochemical conversion device.It produces electricity by converting the chemical energy of a fuel and an oxidant which reacts in the presence of an electrolyte .

Fuel cells differ from ordinary electrochemical cell because here the reactants are supplied from outside rather than being an integral part of the cell and the energy can be withdrawn indefinitely from a fuel cell as long as the outside supply of fuel is maintained.
It is similar to the electricity generator set, which converts the chemical energy of petrol and air into electricity. It is a power generator rather than a rechargeable battery. The most successful and important fuel cell is hydrogen-oxygen fuel cell.

Hydrogen-Oxygen Fuel Cell(acid electrolyte)

The most successful fuel cell is the hydrogen-oxygen fuel cell. It produces electricity based on the reaction: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
It uses hydrogen as the fuel and oxygen as the oxidant.

Construction

*** It consists of two porous graphite electrodes with metal catalysts (like platinum).**
*** Phosphoric acid is used as the electrolyte. Hydrogen and oxygen gas under pressure (50atm) are continuously passed through the anodic and cathodic chambers.**
*** The cell operates at a temperature of about 150-200°C because the ionic conductivity of phosphoric acid is very low at low temp.**

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Heat is generated during the reaction of H2 and O2 so cooling plates are used to maintain the temperature.

The Electrode Reactions are

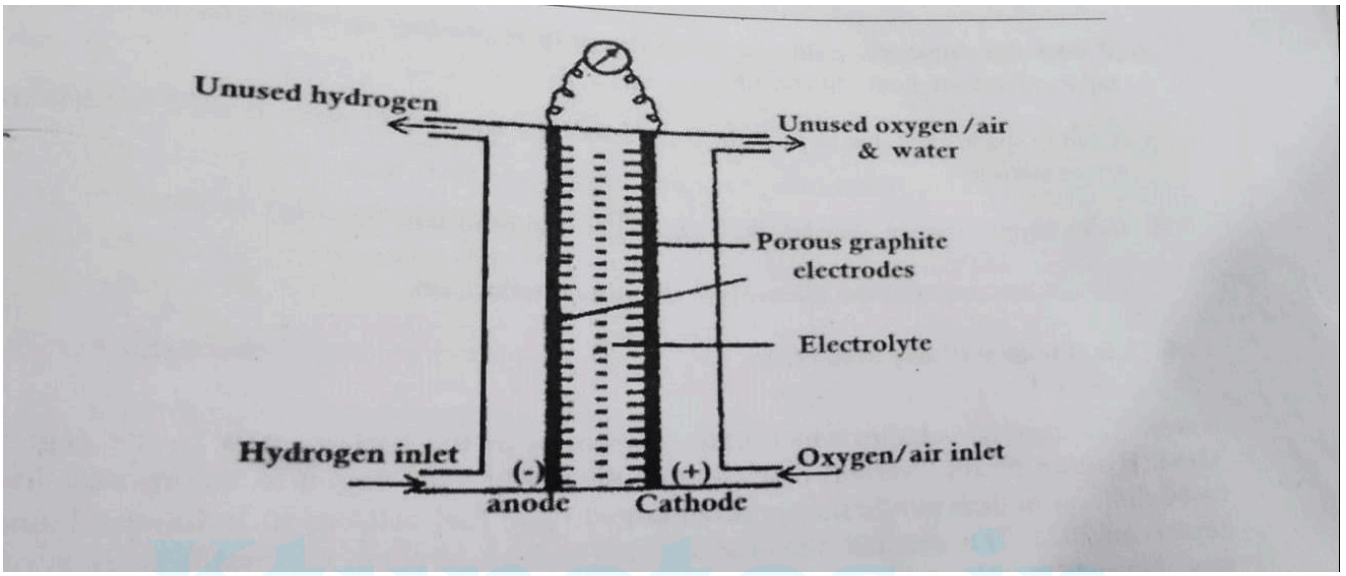
At anode:

$2H_2(g) \rightarrow 4H^+(aq) + 4e^-$ (oxidation of hydrogen)

At cathode:

$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ (reduction of oxygen)

Overall reaction: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$



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Advantages

1. **The fuel cell has high efficiency and can convert about 80% of available chemical energy to electrical energy.**
2. **Environmentally Friendly: The only byproduct is water, making them free from pollution, noise, vibration, and heat transfer.**
3. **Continuous Energy Supply: They can provide continuous energy as long as reactants are supplied.**
4. **Space Applications: Lightweight and the byproduct water is used for drinking in space stations.**
5. **Constant Electrolyte Concentration: The concentration of the electrolyte remains unchanged.**
6. **Simple and Easy to Operate: They are relatively easy to operate.**

Disadvantages of Hydrogen-Oxygen Fuel Cells:

1. **Electro-catalyst Issues: Difficulty in finding a suitable electro-catalyst that functions efficiently for long periods without contamination or deterioration.**
2. **Gaseous Fuel Storage and Handling: Gaseous fuels are difficult to store and handle.**
3. **Corrosive Electrolyte: The electrolyte used is highly corrosive.**
4. **Fuel Purity Requirements: Fuels used must be extremely pure.**
5. **High Cost: The technology is expensive.**

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CONDUCTIVITY

Conductors : A substance or material that allows the passage of electricity, such substances are called conductors.

• Substance or material that does not allow the passage of electricity, are called **non-conductors or insulators.**

Types of conductors

Conductors are two types

1. **Metallic conductor**-Conduct electric current due to movement of electrons
   E.g.: metals, semiconductors.
2. **Electrolytic conductors**-Conduct electric current due to movement of ions .
   Eg: salt solutions.

Conductance(C)

-Reciprocal of resistance

. C=1/R, Units are ohm-1 (mho) and siemen(s).

Specific resistance or resistivity (ρ)

We know that Resistance (R) is directly proportional to length,l

Inversely proportional to area a, i.e **R∝** l/a

R=ρ l /a

Specific resistance (ρ) is the resistance of a conductor of unit length and unit area of cross section. Unit is Ωcm. **ρ= R** a/ l

Specific conductance or conductivity (κ) It is the reciprocal of specific resistance. It is defined as the conductance of 1cm ^{3} of a conductor held between two electrodes of 1cm ^{2} area.

K = 1 /ρ i.e. K = 1 /R * **l**/a

K= C*l/a conductance * l/a

Cell constant It is the ratio of distance between two electrodes (l) and area of the electrodes (a)

Cell constant = l/ aunit - m-1or cm-1

Or K=conductance*cellconstant

Conductivity cell

It consists of two electrode plates coated with Pt black. It is connected to Pt wire, which are sealed through two glass tubes. The tubes are strongly fixed such that distance between electrodes remains unchanged

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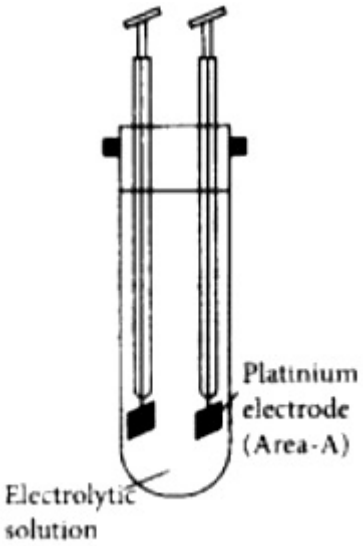

Conductivity Cell

Determination of the Cell constant

Cell constant may be obtained by measuring 'l' and 'a'. But it is very difficult to measure the actual area precisely in this type of small cells. Thus in actual practice cell constant is measured by filling the conductivity cell with a solution of known conductivity like 0.1M KCl solution . The conductivity of 0.1M KCl solution is 0.01288Scm-1 at 298K.

Conductivity of 0.1M KCl solution (K) = Conductance * Cell constant

By measuring the conductance of 0.1M KCl solution in the given cell the cell constant can be determined.

Cell constant = Conductivity of 0.1M KCl solution (K) /Conductance of 0.1M KCl

Measurement of Conductivity with a Digital Conductivity Meter:

* The conductivity cell is connected to the digital conductivity meter and appropriate temperature is selected.
* The conductivity cell is immersed in the sample solution taken in a beaker.Make sure that the cell is fully immersed in the solution .
* The meter measures the conductance by selecting the proper range switch.

The digital conductivity meter directly reads the conductance.

* Conductivity is calculated as: Conductivity = Conductance x Cell constant

3. Problem 1.3.1:

* Calculates the conductivity of a water sample given its conductance (560 µS) and the cell constant determined using 0.1 M KCl (conductivity = 0.01288 S/cm, conductance = 12.34 mS).

Problem 1. Calculate the conductivity of given sample of water at 298K ,which shows a conductance of 560 micro mho in the given cell at 298K.A standard solution of 0.1M KCl shows a conductance of 12.34 mSin that cell.(Given that conductivity of 0.1M KCl at 298 K is 0.001288 ohm-1 cm-1 .

Specific conductance of KCl solution = conductance × cell constant

Cell constant = Sp. conductance of standard KCl solution /conductance of standard KCl in the given 0.01288 /12.34×10−3 = 1.043 cm-1

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Conductivity of given solution = conductance × cell constant = 560×10 -6×1.043 = 583*10 -6 cm-1

2.The specific conductivity of 0.3NKCl at 27^0Cis 0.028SCm^{-1}. The resistance of the cell containing this solution is 300ohm. Determine cell constant?

Given that K=0.028SCm^{-1} R= 300ohm

specific conductivity(K) = 1/R * Cellconstant

Cell constant = K*R = 0.028*300 = 8.4cm^{-1}

3. A conductivity cell is found to have two parallel plates of area 1.5cm^2 kept at 9.8cm apart.It gives a resistance of 1500ohm when filled with solution ? Find out the cell constant and conductivity of the solution?

Cell constant = l/ a = 9.8/1.5 6.53 cm^{-1}

Specific conductance of KCl solution = conductance × cell constant

= 1/1500*6.53 = 4.35*$10^{-3}ohm^{-1}cm^{-1}$

CORROSION

Corrosion is a process which involves the slow destruction of a metal as a result of its reaction with moisture and other gases present in the atmosphere.

Corrosion always occurs at anodic areas. More reactive metals are corroded more easily.

All metals and alloys are susceptible to corrosion. Au shows excellent resistance under atmosphere condition, but gets corroded when exposed to Hg at ambient temperature.

Corrosion is enhanced by the presence of

a) impurities b)air and moisture c)electrolytes and strains in metals like dents, scratches etc.

Corrosion of iron is called rusting.

Rust is a hydrated form of ferricoxide, .

Tarnishing of silver, formation of green coating on copper and bronze etc are other examples of corrosion.

TYPES OF CORROSION

Based on environment corrosion is classified into Two types.

1. Dry or Chemical Corrosion
2. Wet or Electrochemical Corrosion

Dry or Chemical Corrosion

Direct chemical action of environment/atmosphere gases such as O2, halogen, H2S, SO2, N2

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or Corrosion by other gases.

Wet or Electrochemical Corrosion

The corrosion of metals in aqueous environment is more common than dry corrosions. This type of corrosion takes place when

• A metal is in contact with a conducting liquid

• Two dissimilar metals or alloys are immersed or dipped partially in a conducting solution.

According to electrochemical theory

• Separate anodic and cathodic areas are developed, between which current flows through the conducting medium.

• At the anodic area (in the metal part) oxidation and generation of electrons takes place.

• These electrons formed in anodic area moves through the conducting medium, and it is called corrosion current.

• Environment is the cathodic area where the electrons are received(reduction)non-metallic ions like OH- are formed.

• Diffusion towards the metallic ions formed in the anodic area and non-metallic ions in the cathodic area takes place and the corrosion product is formed in between the anodic and cathodic area. .

Mechanism of corrosion (Rusting)

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In this type of corrosion one part of the metal becomes anode, flow of electronic current between the anodic and cathodic areas takes place in through an electrolyte in contact with it.

1)**Hydrogen evolution corrosion** in acidic medium

This Occurs when metals are in contact with acidic solutions in the absence of sufficient oxygen. Eg: Fe or Al in contact with HCl.

Hydrogen evolution corrosion takes place in the case of Iron immersed in acid medium this is due to the fact that hydrogen electrode potential in acid medium (E^0H+/H2)=0V) is higher than electrode potential of iron(E^0 Fe^{2+}/Fe = -0.44V).

Reactions:

Anode $Fe \rightarrow Fe^{2+} + 2e$ $E^0 Fe^{2+}/Fe = -0.44V$

Cathode: $2H^+ + 2e \rightarrow H_2$ $E^0 = 0V$

Net Reaction: $Fe + 2H^+ \rightarrow Fe^{2+} + H_2$ $E^0cell = 0 - (-0.44) = 0.44V$

This type of corrosion occurs when a metal displaces hydrogen ions (H+) from an acidic solution.

Metals with higher negative reduction potentials, like iron (Fe), zinc (Zn), and aluminum (Al), act as the anode and corrode.

Metals with positive reduction potentials, like copper (Cu) and silver (Ag), act as the cathode and

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receive some galvanic protection.

2)Hydrogen Evolution in Alkaline Medium:

Iron doesn't undergo hydrogen evolution corrosion in an alkaline medium because the hydrogen electrode potential in alkaline conditions (-0.828 V) is lower than iron's electrode potential (-0.44 V).

However, metals like aluminum can experience hydrogen evolution in alkaline environments because hydrogen electrode potential in alkaline conditions (-0.828 V) is higher than the electrode potential of Aluminium (-1.66V)

Anode , $Al \rightarrow Al^{3+} + 3e$ $E^0 = -1.66V$

Cathode . $2H_2O + 2e \rightarrow H_2 + 2OH^-$ $E^0 = -0.828V$

Corrosion cell potential is Ecell = Ecathode- Eanode = 0.832V

3) Oxygen Absorption Corrosion in Acidic Medium:

This happens when metals come into contact with water containing dissolved oxygen in acidic or neutral conditions Eg : Rusting of iron.

Reactions:

Anode $Fe \rightarrow Fe^{2+} + 2e$ $E^0 Fe^{2+}/Fe = -0.44V$

Cathode $O_2 + 4H^+ + 4e \rightarrow 2H_2O$ $E^0 = 1.23V$

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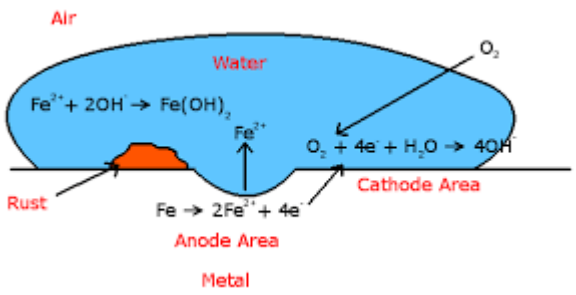
E^0 cell = Ecathode- Eanode = 1.67V

Fe^{2+} produced can be further oxidized to Fe^{3+} in acidic or neutral environments. Fe^{3+} produced forms an insoluble precipitate of Fe(OH)3 which is called yellow rust.

In presence of limited supply of oxygen the corrosion product is the black rust **Fe3O4**(black rust).

The rust formation leads to intense localized corrosion called **pitting corrosion**



4) Oxygen Absorption Corrosion in alkaline Medium:

This happens when metals come into contact with water containing dissolved oxygen in alkaline environment. Eg : Rusting of iron.

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Anode $Fe \rightarrow Fe^{2+} + 2e$ $E^0 Fe^{2+}/Fe = -0.44V$

Cathode: $O_2 + 2H_2O + 4e \rightarrow 4OH^-$ $E^0 = +0.40V$

Here corrosion cell potential in acid medium is:

Ecell= Ecathode- Eanode = 0.40 -(- 0.44) = 0.84V

The variation of potential with pH of oxygen electrode is given by E = 1.23 - 0.0591pH

For alkaline solution pH can be taken as 14 and therefore E = 1.23 - (0.0591 x 14) = +0.40V

Fe^{2+}produced here cannot further oxidized to Fe^{3+}in alkaline environments, since oxygen electrode potential is less than (E^0 Fe^{3+}/Fe^{2+} = +0.77 V) in alkaline medium.

Here no insoluble precipitate of Fe(OH)3 is produced, thus no intense localized corrosion

Galvanic series

Galvanic series which gives real and useful information regarding corrosion behaviour of metals and alloys in a given environment. **It is a series obtained by arranging metals and alloys in the order of corrosion tendencies.**

Mg, Mg alloys, Zn, Al, Cd, Mild Steel, Cast iron, Solder, Pb, Sn, Brass, Cu, Ni, Stainless steel, Cr, Ag, Ti, Au, Pt ——————————— decreasing order of corrosion (Galvanic Series)

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ELECTROCHEMICAL SERIES	GALVANIC SERIES
A series in which the metals and arranged in the increasing order of their Standard Reduction Potential	A series in which the metal electrodes are nonmetals are arranged in the increasing order of their corrosion tendency.
The position of a given metal in electrochemical series is fixed	Position may shift.
$E^0$is measured for metals and non-metals.	No information regarding position of alloys
It tells the relative displacement tendencies.	Predicts relative corrosion tendencies
Electrode potential is measured by using SHE as reference electrode.	Electrode potential is measured by using SCE as reference electrode
$E^0$is measured by dipping pure metals in their 1M salt solution	$E^0$is measured by dipping metals and non-metals in unpolluted sea water

CATHODIC PROTECTION

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Cathodic protection is a method of preventing a metallic object from corrosion. It is an electrochemical process in which the metal which is to be protected is forced to act as cathode.

A metal surface buried or immersed can contain anodic and cathodic areas. Corrosion occurs only in the anodic area. Therefore the entire surface of the metal can be turned into a cathode, then the corrosion can be prevented.

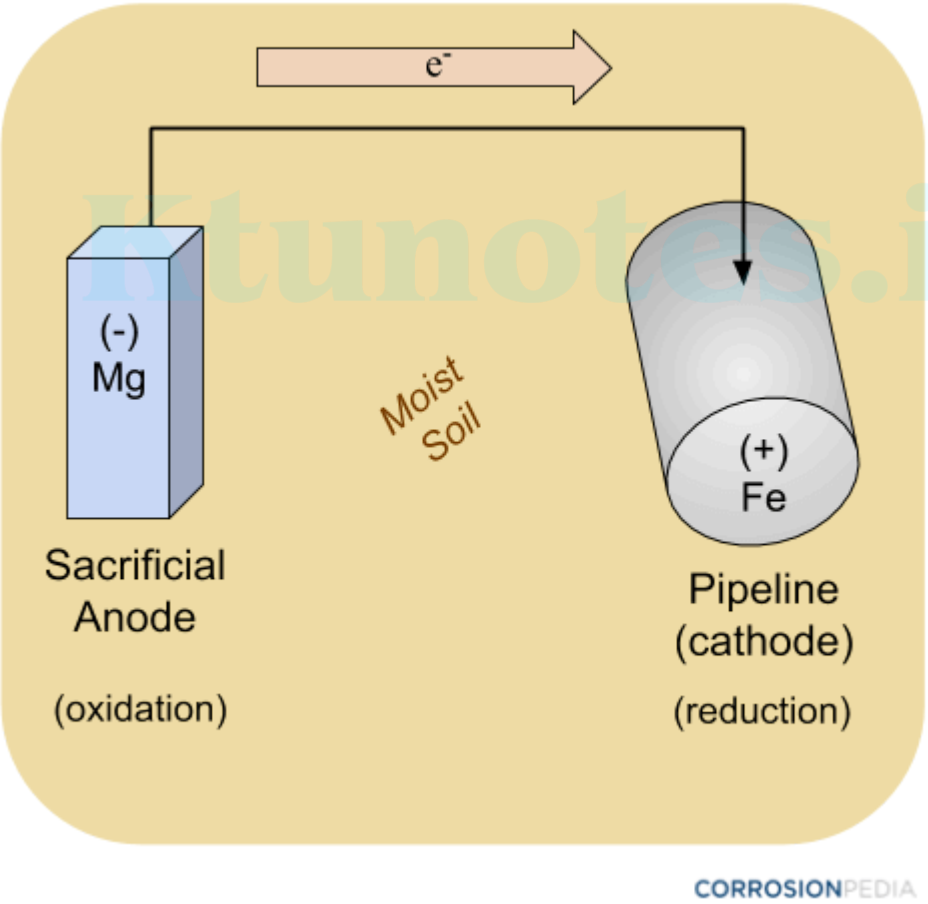
Two main methods of applying cathodic protection are

i. Sacrificial anode protection

ii. Impressed current cathodic protection

(1)Sacrificial anode protection

In this method, the metallic structure which is to be is protected is connected to a more anodic metal through a wire, so the entire corrosion is concentrated on the anodic metal, and the metallic structure is protected sacrificially. Example for anodic metals; Mg ,Zn ,Al and its alloys.



Sacrificial anode systems are based on the position of the metal in the galvanic series. Metals at the top of the series are more prone to corrosion. Sacrificial anode systems are more reliable than the impressed current system because lesser quantity of the component is destroyed and no power supply is required.

Advantages

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1.No external power is required 2.Easy to install and inexpensive 3.Minmum maintenance required

Applications

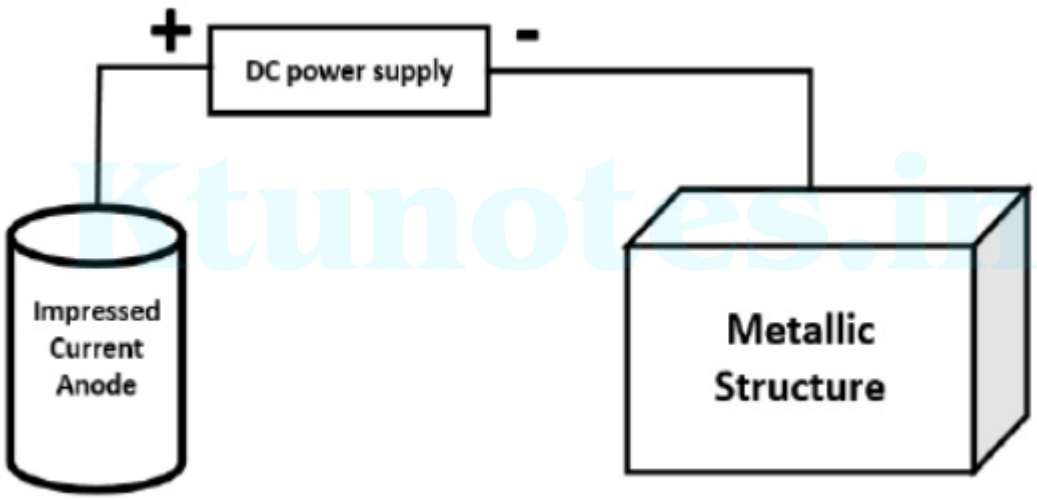
Protection of buried pipelines, underground cables, marine structures, ship body, open water tank coolers, condensers etc.

(2) Impressed current cathodic protection

This widely used system is usually developed to protect a structure which is immersed in a high resistivity electrolyte such as fresh water or sandy soils.

This method requires large amount of current. As a result of corrosion an electric flow occurs from the anodic region to the cathodic part, which is known as corrosion current.

An impressed current is applied in opposite direction to neutralize corrosion current and convert the corroding metal from anode to cathode. Usually impressed current is derived from a battery or DC source with an insoluble anode like graphite or silica



Applications

Impressed current cathodic protection is applied to

(i)Open water boxes, coolers, water tanks etc.

(ii)Concrete buildings, bridges etc.

(iii)Vessels, pipelines and tanks which are used to store or transport liquids. (iv)Marine cathodic protection covers many areas jetties, harbours, offshore structures etc.

Electroplating

Electroplating is a process by which the coating metal is deposited on the base metal by

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passing a direct current through an electrolytic solution containing the soluble salt of the coating metal

Theory of electroplating

The object to be plated is the cathode of the electrolytic cell and the anode is either the plating metal or an inert material of good electrical conductivity like graphite.The electrolyte is a soluble salt solution of the coating metal .

When DC current is passed through the electrolyte electrolysis occurs and the metal is coated on the surface of the solution.

Objectives of Electroplating:

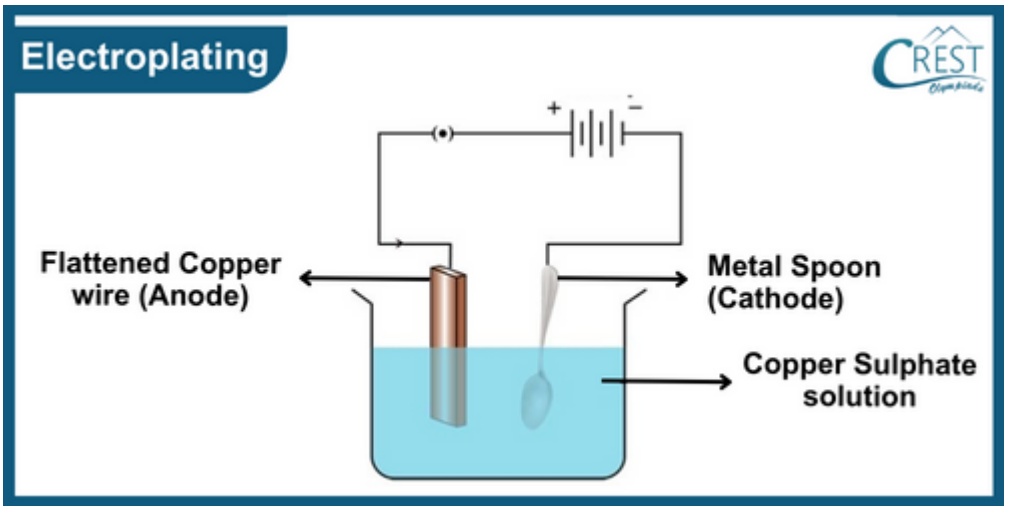
- **To improve corrosion resistance**
- **To increase chemical attack and wear resistance**
- **To enhance physical appearance and hardness**
- **To improve surface properties**
- **To increase decorative and commercial value**

Copper Electroplating:

The article to be plated is cleaned to remove oil, grease, and surface impurities.It is then treated with dilute acid (like HCl or H2SO4) to remove oxides and scales. The cleaned article is then made as cathode of the electrolytic cell.Pure copper metal rode is taken as anode. Both the electrodes are dipped in copper sulphate solution. DC current is passed through the electrolyte electrolysis occurs and Cu^{2+} move towards the cathode and gets deposited there.

Anode $Cu \rightarrow Cu^{2+} + 2e$

Cathode $Cu^{2+} + 2e \rightarrow Cu$



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f the anode is made of some inert material like graphite, then electrolytic salt is added continuously in order to maintain proper coating metal ion concentration in the electrolytic bath.

Moreover, the process of electroplating goes on non-stop, since the anode is not consumed and its replacement is not required. When inert electrode is used, anode reaction can be represented as

$$H_2O \rightarrow 2H^+ + 1/2\ O_2 + 2e$$

For brighter and smooth deposits, favorable conditions such as low temperature, medium current density and low metal ion concentration are used.

Applications of copper electroplating

1. **Steel articles used in automobile industry are generally given three successive coatings of Cu, Ni and Cr. This is done for corrosion protection and to impart shining and clear metal appearance.**
2. **Copper electroplating is used in decorative items, kitchen wares, jewelry etc. to enhance the aesthetic appearance.**

Electroless plating

Electroless plating is a chemical, autocatalytic, metal coating process without the use of electricity.

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Principle

In this process metal piece, which is to be plated is immersed in a mixture of a reducing agent and a complex compound. It produces a deposition of continuous coating layer, on the surface, by a controlled chemical reduction that is catalysed by the metal or alloy being deposited

Metal ions + Reducing agent→Metal plated over a catalytically active surface + oxidised product.

Electroless plating is used to deposit a conductive surface on a non- conductive object to improve its electroplating

Process of Electroless plating

Process of Electroless plating involves the following steps

1. **Preparation of the surface to be plated**: In this step the surface is activated by,

   The surface of the metal which is to be washed thoroughly,
   a) Etching: Removal of impurities by acid treatment
   b) Electroplating: A thin layer of the metal to be plated or any other suitable metal is coated on the surface of the object
   c) Chemical treatment: The plastic objects and printed circuit boards are treated with $SnCl_2$ and $PdCl_2$

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2. Preparation of Plating bath : The plating bath should include the following

a) A soluble salt of the metal (chlorides and sulphates)
b) Complexing Agents like EDTA, tartrate citrate etc to prevent excess deposition
c) Reducing agents like Sodium thio Sulphate, Formaldehyde etc
d) Accelerators like succinates, glycinates, fluorides etc to increase the plating rate.
e) Stabilizers like Calcium ions, lead ions, thio urea etc to prevent decomposition of the bath and to impart stability to the solution
f) Buffer solution like Rochelle salt, Boric acid to maintain pH

3. Procedure:
The article to be plated is immersed in the bath containing the metal salt and reducing agent. The plating is carried out in a series of tanks .The rate of deposition is controlled by the amount of reducing agent and the type of complexing agent

Advantages

1.No electricity required.
2.Can be used to obtain uniform coating on irregular shaped articles.
3.The deposits are more compact and highly adherent.
4.No elaborate arrangements are required.
5.Coating provides better wear resistance
6. The coating is less porous and provides better corrosion resistance.
7. Improved hardness, strength, ductility

Electroless -Copper plating.

Electroless copper plating is used to deposit a layer of copper on substrate without the use of electricity.

The object which is to be plated is degreased using organic solvents followed by acid treatment .This object is then immersed in plating bath containing cupric salt ,reducing agents,buffer and complexing agent.

Soluble salt is the Copper sulphate, formaldehyde is the reducing agent a. buffer (NaOH)complexing agent EDTA is taken in the plating bath. Metal surface(metal)which is to be plated is introduced into the bath and then heated. The electroless plating of copper take place as follows.

At cathode

$Cu^{2+}+2e- \rightarrow Cu$

At anode

$2HCHO+4OH^{-} \rightarrow 2HCOO- + 2H_2O+H_2+2e$

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Over all reaction is

$Cu^{2+}+2HCHO+4OH^- \rightarrow Cu + 2\ HCOO^- + 2H_2O + H_2$

Applications of electroless copper plating

1. **This type of coating is mainly used in double or multilayered printed circuit boards in which plating through holes is required.**
2. **This is used as base coating for subsequent electroplating.**
3. Numerous consumer goods are coated by this method to create durable and attractive surfaces.

Limitations

1.The complexing agent creates difficulties.

2.Formaldehyde is a human health hazard.

1. Calculate the emf for the cell, Zn/Zn+ // Ag+ /Ag given E^0 Zn 2+ / Zn = -0.762v and E^0Ag+/Ag = 0.8 v
   A: Given cell is Zn/Zn^{2+}//Ag+ /Ag. E 0 Zn2+/Zn =- 0.762 v E 0 Ag+/Ag =0.8 v E 0 cell = E0 right – E 0 left = 0.8 – (-0.762) = 1.562 v
   Applying Nernst eqn

   $E_{cell} = E^0cell – 0.0591/n \log [p]/[R]$ n=2

2. Calculate the E0 cu 2+/cu, given E^0cu 2+ /cu = 0.296 v and [cu+2] = 0.015M E =

   $E^0 + 0.0591/2 \log 0.296$

3. Write the half cell and net cell reactions for the following cell, Zn / Znso4 (aq) // cuso4 (aq) / cu. Calculate the standard emf of the cell given, E 0 Zn +2 /Zn =- 0.76 v and E0 cu+2/cu = + 0.34 v.
   A: Half cell reactions At anode:
   Zn Zn+2 + Ze
   At cathode:
   cu+2 + e- cu.
   Net cell reaction = Zn + cu+2 Zn+2 + cu.
   E 0 cell = E0 Cathode – E 0 Anode. = E0 cu +2 /cu – E 0 Zn +2 / Zn = 0.34 – (-0.76) = 1.1 v.

UNIVERSITY QUESTIONS

Calculate the equilibrium constant for the following reaction at 250C Fe(s) + Cu^{2+}(aq) = Fe $^{2+}$(aq) + Cu (s) Given E^0 Fe^{2+}/Fe= -0,44 V, E0Cu^{2+}/Cu = 0.34 V.

Log K= nE^0cell / 0.0591 E^0cell = 0.34- (-0.44) = 0.78V

2* 0.78/0.0591 = 26.39 K= $\log^{-1}$26.39

2 . Give the electrochemical reaction taking place when an iron nail is dipped in dil.HCl. E^0Fe^{2+}/Fe= -0,44 V, E0Fe3+/Fe= -0,04 V, E0H+/H2= 0 V.

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When an iron nail is dipped in HCl two cells are developed. (Fe2+ and Fe 3+)

Emf of the first cell (Fe2+/Fe//H+/H) = $E_{cathode} - E_{anode}$ = 0-(-0.44) = 0.44V Emf

of the second cell (Fe3+/Fe/ H+/H) = 0- (-0.04) = 0.04V

The emf of the first cell is higher so the reaction takes place in Fe2+

So Fe^{2+} +2 HCl ——————— $FeCl_2$ + H2

4. Why Mg corrodes in both acidic and alkaline oxygen deficient conditions, whereas Fe does not corrode in alkaline oxygen deficient condition? Mg2++ 2e→ Mg, E0 = -2.36 V, Fe2+ +2e→Fe, E0= -0.44 V, H+ +e→1⁄2H2, E0=0

   The electrode potential of H2 electrode (E) = -0.0591pH (O2 deficient)
   When pH = 0 , E = 0 (acidic condition)
   When pH = 14 E = - 0.0591*14 = -0.8274V
   In the case of Mg , both in acid and alkaline condition E0 of Mg(-2.36) is less than the electrode potential of H2. So Mg corrode in both acidic and alkaline conditions. But in the case of Fe , in alkaline condition the E0 of Fe (-0.44V) is greater than E0 of H2 (-0.82V) so Fe do not corrode in alkaline condition.

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